A Codimension-One Entropy Refinement for Dittert's Conjecture in Dimension Seven

Extension note to the dimensions 8–15 manuscript

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Abstract

This note gives a proposed extension of the Hall-deficit proof of Dittert's conjecture from dimensions 8–15 to dimension 7. The new point is a geometric-mean lower bound for the cofactors of a selected column of a doubly stochastic matrix. Applied twice, it strengthens the sharp permanent bound for a zero rectangle of codimension one: besides the sharp constant, the bound retains the entropy of both marginal distributions of the critical core. The only two-sided Hall configurations not covered by the existing scalar method in dimension 7 have codimension one. Critical-core stationarity and a conditional product-energy estimate then reduce those configurations to three exact bivariate polynomial inequalities. Their positivity on a rational square is certified by exact Bernstein subdivision. The remaining Hall configurations are covered by the original second-order and one-sided arguments with sharpened constants. No floating-point assertion is used.

1 Statement and inherited notation

For a nonnegative $n \times n$ matrix A whose entries sum to n, write r_i and c_j for its row and column sums and put

$$\Phi(A) = \prod_i r_i + \prod_j c_j - \mathrm{per}(A), \qquad \gamma_n = \frac{n!}{n^n}.$$

We use the arbitrary-dimensional parts of the accompanying manuscript: positive global maximizers are uniform; a hypothetical boundary maximizer has deficits

$$P := \prod_i r_i = 1 - \rho, \qquad Q := \prod_j c_j = 1 - \sigma, \qquad H := \mathrm{per}(A) = \gamma_n - \delta,$$

with

$$0 \le \rho, \sigma, \qquad \rho + \sigma \le \delta \le \gamma_n; \tag{1.1}$$

and a critical Hall pair produces a dilation $S = A/(1-\tau)$ dominating a doubly stochastic matrix B with a forced zero rectangle.

We also use

$$v_k = \frac{k!}{k^k} \quad (k \ge 1), \qquad v_0 = 1,$$

and the sharp zero-rectangle bound

$$\mathrm{per}(B) \ge L^\sharp_{n,a,b} = \frac{v_{n-a} v_{n-b}}{v_{n-a-b}}. \tag{1.2}$$

The purpose of the note is to prove the following extension.

Theorem 1.1. *For every nonnegative 7×7 matrix A whose entries sum to 7,*

$$\Phi(A) \le 2 - \frac{7!}{7^7}.$$

Equality holds only at $A = J_7/7$.

The interior classification is already dimension-free. It therefore suffices to exclude a boundary global maximizer when $n = 7$.

2 The new permanent inequality

The missing information in the sharp bound (1.2) is the nonuniformity of the mass that crosses the critical core. The following cofactor inequality recovers it.

Lemma 2.1 (Geometric cofactor entropy)**.** *Let M be an $m \times m$ doubly stochastic matrix, let*

$$\xi = (\xi_1, \dots, \xi_m)$$

be one selected column, and let

$$d_i = \operatorname{per} M(i \mid \xi)$$

be the permanent of the submatrix obtained by deleting row i and the selected column. Then

$$\prod_{i=1}^{m} d_i^{\xi_i} \ge v_m \exp\left\{ \frac{1}{2} \sum_{i=1}^{m} \left(\xi_i - \frac{1}{m} \right)^2 \right\}. \tag{2.1}$$

The usual convention is used for zero weights, and the boundary cases follow by continuity.

Proof. Write Y for the other $m-1$ columns and, initially assuming $0 \le \xi_i < 1$, normalize its rows by

$$n_{ij} = \frac{y_{ij}}{1 - \xi_i}.$$

Let

$$L_i(x) = \sum_{j=1}^{m-1} n_{ij} x_j, \qquad p_i = \operatorname{per} N_{-i}.$$

Then

$$d_i = \left(\prod_{k \ne i} (1 - \xi_k) \right) p_i. \tag{2.2}$$

For positive parameters t_i, consider

$$q_t(x) = \sum_{i=1}^{m} \xi_i t_i \prod_{k \ne i} L_k(x). \tag{2.3}$$

It is real stable, because it is obtained by differentiating in z at $z = 0$ the stable polynomial

$$\prod_{i=1}^{m} \big(L_i(x) + \xi_i t_i z \big).$$

Weighted AM–GM first gives

$$q_t(x) \geq \left(\prod_i t_i^{\xi_i}\right) \prod_k L_k(x)^{1-\xi_k}.$$

A second weighted AM–GM inequality gives

$$L_k(x) \geq \prod_j x_j^{n_{kj}}.$$

Since $(1-\xi_k)n_{kj} = y_{kj}$ and every column of Y sums to one,

$$q_t(x) \geq \left(\prod_i t_i^{\xi_i}\right) x_1 \cdots x_{m-1}.$$

Thus

$$\operatorname{Cap}(q_t) \geq \prod_i t_i^{\xi_i}.$$

Square-free coefficient extraction for stable polynomials yields

$$\sum_i \xi_i t_i p_i \geq v_{m-1} \prod_i t_i^{\xi_i}. \tag{2.4}$$

Putting $t_i = p_i^{-1}$, with the zero-weight terms omitted and then using continuity, gives

$$\prod_i p_i^{\xi_i} \geq v_{m-1}. \tag{2.5}$$

Taking the ξ-weighted geometric mean in (2.2),

$$\prod_i d_i^{\xi_i} \geq v_{m-1} \prod_k (1-\xi_k)^{1-\xi_k}. \tag{2.6}$$

For $h(t) = (1-t)\log(1-t)$ one has $h''(t) = 1/(1-t) \geq 1$. Hence

$$\sum_k h(\xi_k) \geq mh(1/m) + \frac{1}{2}\sum_k (\xi_k - 1/m)^2.$$

Finally,

$$v_{m-1} \exp\{mh(1/m)\} = v_{m-1}\left(1 - \frac{1}{m}\right)^{m-1} = v_m,$$

and (2.1) follows. □

The codimension-one case of the critical zero rectangle now has a two-sided entropy correction.

Theorem 2.2 (Codimension-one block entropy). *Let B be an $n \times n$ doubly stochastic matrix with a zero block*

$$B[R, C] = 0, \qquad |R| = a, \qquad |C| = b, \qquad a + b = n - 1.$$

Put $I = R^c$ and $J = C^c$. The core $X = B[I,J]$ has total mass one. Let ξ and η be its row and column marginals and put

$$s_I = \sum_{i\in I}\left(\xi_i - \frac{1}{b+1}\right)^2, \qquad s_J = \sum_{j\in J}\left(\eta_j - \frac{1}{a+1}\right)^2.$$

Then

$$\operatorname{per}(B) \ge v_{b+1}v_{a+1}\exp\left(\frac{s_I + s_J}{2}\right). \tag{2.7}$$

At $s_I = s_J = 0$, the constant is exactly the sharp bound (1.2) *with $p = 1$.*

Proof. Write $Y = B[I,C]$ and $Z = B[R,J]$. Every perfect matching uses exactly one core edge. If

$$d_i = \operatorname{per} Y_{-i}, \qquad e_j = \operatorname{per} Z_{-j},$$

then

$$\operatorname{per}(B) = \sum_{i\in I,\, j\in J} x_{ij} d_i e_j.$$

Weighted AM–GM, using the core entries x_{ij} as weights, gives

$$\operatorname{per}(B) \ge \prod_{i\in I} d_i^{\xi_i} \prod_{j\in J} e_j^{\eta_j}. \tag{2.8}$$

The square matrix obtained by adjoining ξ to Y is doubly stochastic of order $b+1$. Lemma 2.1 therefore bounds the first product in (2.8) by $v_{b+1}\mathrm{e}^{s_I/2}$. Likewise, adjoining η^T as the last row of Z and transposing bounds the second product by $v_{a+1}\mathrm{e}^{s_J/2}$. $\square$

3 Dimension-seven constants and the routine Hall pairs

For the rest of the proof let

$$n = 7, \qquad \gamma = \gamma_7 = \frac{720}{117649}, \qquad \eta = \frac{1}{47}.$$

Define $\alpha_0 = 0$ and, for $1 \le k \le 6$,

$$\alpha_k = \begin{cases} 14\left(\frac{k}{7} + \eta\right)\left(1 - \frac{k}{7} - \eta\right), & 2k < 7, \\ \frac{2k(7-k)}{7}, & 2k \ge 7. \end{cases} \tag{3.1}$$

The relative-entropy subset proof from the main manuscript applies verbatim, because

$$\eta^2 - \frac{\gamma}{14(1-\gamma)} = \frac{23263}{1808073127} > 0, \qquad \frac{3}{7} + \eta < \frac{1}{2}, \tag{3.2}$$

and $\alpha_k \le 7/2$ for every k.

Let $a = |I^c|$, $b = |J^c|$, and $p = 7 - a - b$. For two-sided pairs with $p \ge 2$, use

$$T = \frac{5}{24}, \qquad \lambda = \binom{7}{2} - \binom{7}{3}T = \frac{329}{24}.$$

The Hall estimate gives $\tau < T$, since

$$T^2 - \frac{7\gamma}{1-\gamma} = \frac{20185}{67351104} > 0.$$

The same alternating-binomial argument gives

$$(1-t)^7 \geq 1 - 7t + \frac{329}{24}t^2 \qquad (0 \leq t \leq 5/24).$$

Substitution in the second-order scalar margin, with

$$L = L^{\sharp}_{7,a,b}, \qquad c = \frac{\alpha_a + \alpha_b}{p^2},$$

shows that all six pairs with $p \geq 2$ have positive margin. The smallest is the pair $(a, b) = (1, 1)$:

$$\frac{217499955444827479495}{21713873952987016007066496} > 0. \tag{3.3}$$

It remains to discuss $p = 1$. Up to transposition, the pairs are

$$(a, b) = (1, 5), (2, 4), (3, 3). \tag{3.4}$$

Write

$$e = \sum_{i \in I^c} r_i - a, \qquad f = \sum_{j \in J^c} c_j - b, \qquad x = s_A(I^c, J^c).$$

Then

$$\tau = e + f - x. \tag{3.5}$$

If $e \leq 0$ or $f \leq 0$, only the positive excess is needed in the Hall estimate. With

$$T = \frac{3}{20}, \qquad \lambda = \binom{7}{2} - \binom{7}{3}T = \frac{63}{4},$$

the six resulting second-order margins are positive. Their minimum is

$$\frac{165035926385}{3897251304526692} > 0. \tag{3.6}$$

Thus the only new case is (3.4) with

$$e > 0, \qquad f > 0. \tag{3.7}$$

The subset bounds imply separately

$$0 < e, f < \frac{3}{20}. \tag{3.8}$$

4 One-sided Hall cuts in dimension seven

For completeness, the one-sided part of the original proof also extends to $n = 7$ once its two coarse numerical replacements are removed.

Put

$$d_0^2 = \frac{7\gamma}{2(1-\gamma)}.$$

Then $d_0 < 3/20$, and every row and column sum is at least 17/20. Moreover

$$\frac{(1-\gamma)(17/20)^2}{2} > \frac{1}{3}. \tag{4.1}$$

Repeating the row-product energy proof therefore gives

$$\rho > \frac{1}{3}\sum_i \left(1 - \frac{1}{r_i}\right)^2. \tag{4.2}$$

The column analogue is identical.

In the small-entropy regime, the exact certificate replacing the coefficient 2/5 is

$$\frac{1}{3}(1-\gamma)^2\gamma - \frac{49}{2}\kappa_7^2 = \frac{221764894862625438 50191055}{2067270180525557448014 4334848} > 0, \tag{4.3}$$

where $\kappa_7 = v_6 G(6) = 15625/2519424$.

For the large-entropy regime, retain the exact dependence on τ instead of replacing $(1-\tau)^{7/2}$ by a fixed decimal. It is enough to prove, for $0 \le t \le 3/20$,

$$\frac{2}{3}(1-\gamma)^2(1-t)^6 - \gamma(7 - 21t + 35t^2)^2 > 0. \tag{4.4}$$

Every Bernstein coefficient of the polynomial in (4.4) on $[0, 3/20]$ is positive; the minimum is

$$\frac{155130361249919329}{1328763571296000000} > 0. \tag{4.5}$$

Indeed, the exact one-sided inequality before squaring is

$$(1-\gamma)\sqrt{\frac{2\gamma}{3}}(1-\tau)^{5/2} > \gamma\frac{1-(1-\tau)^7}{\tau}.$$

On this interval,

$$(1-\tau)^{5/2} \ge (1-\tau)^3, \qquad \frac{1-(1-\tau)^7}{\tau} \le 7 - 21\tau + 35\tau^2,$$

which reduces the claim to (4.4). Consequently all one-sided Hall cuts are excluded.

5 The hard codimension-one configurations

We now assume (3.4), (3.7), and (3.8). Put

$$R = I^c, \quad C = J^c, \quad m = |I| = 7 - a = b + 1, \quad \ell = |J| = 7 - b = a + 1, \quad d = \min\{a, b\}.$$

5.1 Core saturation and stationarity

Because $p = 1$, both $B[I, J]$ and $S[I, J]$ have total mass one. Since $B \le S$, one has entrywise saturation

$$B[I, J] = S[I, J] = \frac{A[I, J]}{1 - \tau}. \tag{5.1}$$

Let ξ and η be the row and column marginals of this core and put

$$s = s_I + s_J. \tag{5.2}$$

For a permutation, the number of selected entries in $I \times J$ exceeds the number selected in $R \times C$ by exactly one. Let H_k be the total permanent weight of permutations using exactly k entries of $R \times C$, so that

$$H = \sum_{k=0}^{d} H_k.$$

Scale the critical core $A[I, J]$ by a parameter and globally renormalize to keep total mass seven. Stationarity at the maximizer gives the exact identity

$$P\left(\sum_{i\in I} \frac{\xi_i}{r_i} - 1\right) + Q\left(\sum_{j\in J} \frac{\eta_j}{c_j} - 1\right) = \frac{\tau H + \sum_{k=1}^{d} kH_k}{1 - \tau}. \tag{5.3}$$

Indeed, differentiating the permanent contributes $\sum_k (k + 1)H_k$, while global normalization contributes $-(1 - \tau)\Phi(A)$.

Let H_0 now denote the contribution of the permutations avoiding $R \times C$. Theorem 2.2, (5.1), and $\tau \le e + f$ give

$$H_0 \ge L(1 - \tau)^7 \mathrm{e}^{s/2} \ge L(1 - e - f)^7 \mathrm{e}^{s/2}, \qquad L = v_{b+1}v_{a+1}. \tag{5.4}$$

Since $\sum_{k\ge1} kH_k \le d(H - H_0)$, the right side of (5.3) is at most

$$\frac{(e + f + d)H - dH_0}{1 - e - f}. \tag{5.5}$$

5.2 Conditional product energy

Define the groupwise AM–GM envelopes

$$P_a(e) = \left(1 + \frac{e}{a}\right)^a \left(1 - \frac{e}{m}\right)^m, \qquad P_b(f) = \left(1 + \frac{f}{b}\right)^b \left(1 - \frac{f}{\ell}\right)^\ell. \tag{5.6}$$

Then $P \le P_a(e)$ and $Q \le P_b(f)$. Put

$$u_r = P_a(e) - P, \qquad u_c = P_b(f) - Q.$$

Lemma 5.1 (Conditional inverse-energy)**.** *On the box* (3.8),

$$\sum_{i\in I} \left(\frac{1}{r_i} - \frac{1}{m}\sum_{h\in I} \frac{1}{r_h}\right)^2 \le 3u_r, \tag{5.7}$$

and the analogous column inequality is bounded by $3u_c$*.*

Proof. Every row and column sum is at least 17/20. Set $q = 1 - e/m$, so $q \geq 17/20$. For $r, q \geq 17/20$ one has

$$\frac{r}{q} - 1 - \log \frac{r}{q} \geq \frac{(17/20)^2}{2} \left(\frac{1}{r} - \frac{1}{q} \right)^2. \tag{5.8}$$

This follows by putting $y = 1 - q/r$ and writing the left side as

$$\int_0^y \frac{t}{(1-t)^2}\, dt.$$

The appropriate groupwise linear terms cancel. Hence

$$\log \frac{P_a(e)}{P} \geq \frac{(17/20)^2}{2} \sum_{i \in I} \left(\frac{1}{r_i} - \frac{1}{q} \right)^2 \geq \frac{(17/20)^2}{2} V_I,$$

where V_I denotes the left side of (5.7). Since

$$\begin{aligned} P_a - P &\geq P \log \frac{P_a}{P} \\ &\geq (1-\gamma) \log \frac{P_a}{P} \\ &> \frac{1}{3} V_I \end{aligned}$$

by (4.1), the result follows. The column proof is the same. □

5.3 Reduction to a scalar inequality

Put

$$t = e + f, \qquad D = 2 - P_a(e) - P_b(f), \qquad K = \gamma - D, \qquad u = \delta - D. \tag{5.9}$$

The joint budget (1.1) gives

$$u \geq u_r + u_c \geq 0, \qquad H = K - u. \tag{5.10}$$

Also put

$$A_0 = L(1-t)^7, \qquad E = \frac{e}{m-e}, \qquad F = \frac{f}{\ell - f}. \tag{5.11}$$

By the harmonic-mean inequality and Lemma 5.1, the left side of (5.3) is at least

$$P_a(e)E + P_b(f)F - u_r E - u_c F - \sqrt{3su}. \tag{5.12}$$

On the other hand, (5.5), (5.4), and (5.10) give the upper bound

$$\frac{(t+d)K - dA_0}{1-t} - \frac{(t+d)u + dA_0(\mathrm{e}^{s/2} - 1)}{1-t}. \tag{5.13}$$

Because

$$E, F \leq \frac{t+d}{1-t} \qquad \text{and} \qquad u_r + u_c \leq u,$$

the linear product-loss terms in (5.12) are absorbed by the negative u term in (5.13). Thus, with

$$\mathcal{B} = P_a(e)E + P_b(f)F - \frac{(t+d)K - dA_0}{1-t}, \tag{5.14}$$

stationarity requires

$$\mathcal{B} \leq \sqrt{3su}. \tag{5.15}$$

In particular, if $\mathcal{B} > 0$, then

$$su \geq \frac{\mathcal{B}^2}{3}. \tag{5.16}$$

The permanent estimate (5.4) simultaneously gives

$$u + A_0 \mathrm{e}^{s/2} \leq K. \tag{5.17}$$

If $A_0 > K$, this is already impossible. Otherwise, using $\mathrm{e}^{s/2} \geq 1 + s/2$ and AM–GM,

$$su \leq \frac{(K - A_0)^2}{2A_0}. \tag{5.18}$$

Consequently, a contradiction follows whenever

$$\mathcal{B} > 0 \qquad \text{and} \qquad 2A_0\mathcal{B}^2 > 3(K - A_0)^2. \tag{5.19}$$

5.4 Exact Bernstein certificate

All quantities in (5.19) are rational functions of e, f. Set

$$Q_0 = (m - e)(\ell - f)(1 - e - f), \qquad \widetilde{\mathcal{B}} = Q_0\mathcal{B},$$

and

$$\mathcal{C} = 2A_0\widetilde{\mathcal{B}}^2 - 3(K - A_0)^2 Q_0^2. \tag{5.20}$$

The functions $A_0 - K$, $\widetilde{\mathcal{B}}$, and $\mathcal{C}$ are bivariate polynomials with rational coefficients.

Exact Bernstein subdivision on

$$0 \leq e, f \leq \frac{3}{20}$$

proves, for each pair in (3.4), that every point satisfies one of the following alternatives:

$$K < 0, \qquad A_0 - K > 0, \qquad \widetilde{\mathcal{B}} > 0 \ \text{ and } \ \mathcal{C} > 0. \tag{5.21}$$

The first alternative contradicts $H = K - u \geq 0$; the second contradicts (5.17); and the third is (5.19). The exact subdivision audit is

(5.22)

(a, b)	tree nodes	$A_0 - K$ leaves	$\widetilde{\mathcal{B}}, \mathcal{C}$ leaves
$(1, 5)$	39	10	10
$(2, 4)$	35	9	9
$(3, 3)$	47	11	13

A positive Bernstein coefficient lower-bounds the polynomial on its box, and every subdivision is the exact rational de Casteljau transformation. Thus (5.21) is a continuum certificate, not a mesh computation.

This excludes the last three critical Hall configurations.

6 Completion

A hypothetical boundary maximizer has $\tau > 0$. Two-sided pairs with $p \geq 2$ are excluded by (3.3); codimension-one pairs with a nonpositive excess are excluded by (3.6); the remaining codimension-one pairs are excluded by Sections 5.1–5.4; and one-sided pairs are excluded by Section 4. Hence no boundary global maximizer exists in dimension seven. The dimension-free interior classification then implies Theorem 1.1.

Remark 6.1. The conceptual gain is not merely a better numerical constant. The sharp block bound records only the sizes (a, b, p). At $p = 1$, the critical core is itself a probability coupling, and its two marginal deviations are precisely the variables that couple to the stationarity equations. Theorem 2.2 keeps those variables rather than discarding them. That is why the formerly negative scalar margins become coercive after stationarity is imposed.